



XMBA 201A
Economic Analysis for Business Decisions

Price Discrimination

Lecture 5

Professor Steve Tadelis

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Yesterday

- Optimal quantity: solve $MR(q) = MC(q)$
- Optimal Price: plug optimal quantity into the demand function $p(q)$
- FC and AC only relevant for shut down and NOT optimal q / p !
- Never be on the inelastic part of the demand curve (beware short-vs-long run decisions, e.g., customer loyalty and attrition)
- Aggregating demand:
 - add up $q(p)$ quantities across all relevant markets (notice relevant price regions!)
 - Invert demand to $p(q)$ with relevant quantity regions
 - Derive $MR(q)$ from $p(q)$ with relevant quantity regions
 - Equate $MR(q^*) = MC(q^*)$ then get optimal price from demand $p^*(q^*)$
- Group price discrimination

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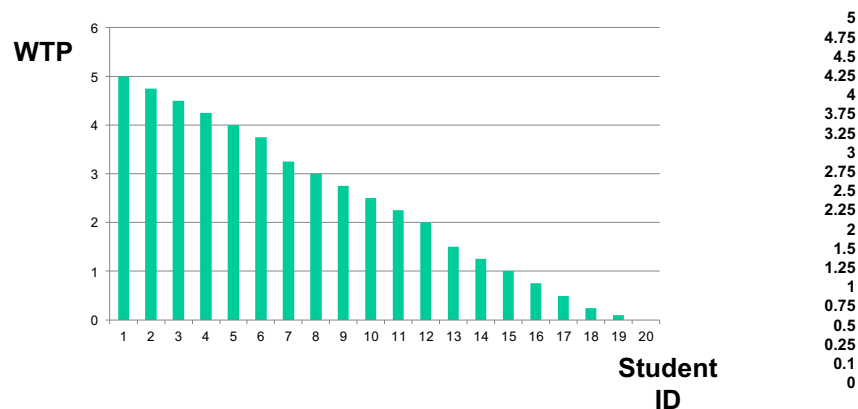
On Orbitz, Mac Users Steered To Pricier Hotels

1. Do you think Orbitz has information to estimate price elasticity of demand for hotels by Mac and PC users? If so, can the company use this information to target price promotions to only PC users?
2. Do you think that Orbitz's price discrimination strategy may be helping or harming consumers?

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Squeezing More Juice from the Market

- **"My Dream"**: charge every consumer what s/he is willing to pay!
- Recall the willingness to pay schedule for chocolate bars from Lecture 5:



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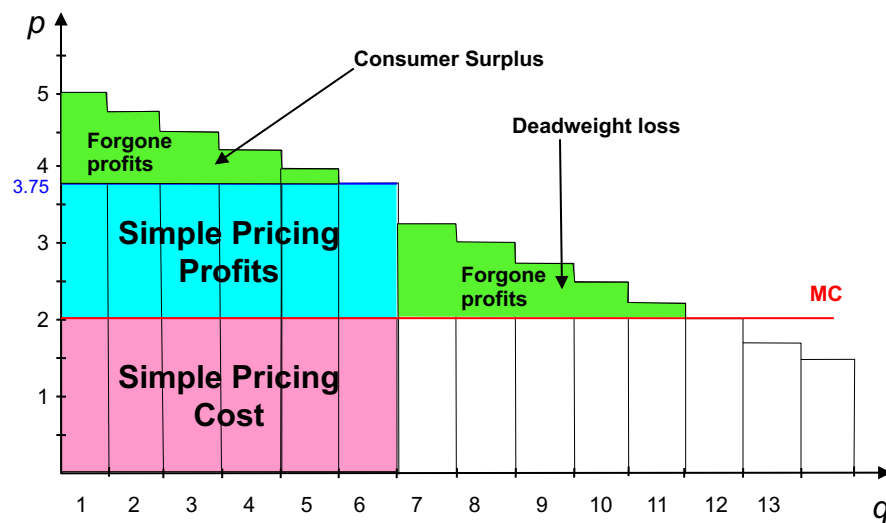
Personalized Pricing (1st Degree PD)

- If, say, $MC = 1$ for each bar of chocolate, then sell to the first 14 students for **individual prices**: $p_1 = 5$, $p_2 = 4.75$, $p_3 = 4.5$, ..., $p_{14} = 1.25$ (we can sell a 15th at a price of \$1, a wash)
- Profits:
 $(5 - 1) + (4.75 - 1) + \dots + (1.25 - 1) = \30.75
- This is almost double the profits from simple pricing (with $p = 3.75$, $q = 6$ and profits of \$10.5 like we saw in Lecture 3).
- **What is simple pricing giving up?**

5
4.75
4.5
4.25
4
3.75
3.25
3
2.75
2.5
2.25
2
1.5
1.25
1
0.75
0.5
0.25
0.1
0

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Consumer Surplus and Deadweight Loss



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Sophisticated Pricing

- **Multi-part tariffs:** Price has several components: fixed fee and per unit
 - Cell-phone fixed fee + data overage;
 - U-Haul rentals: daily rate + miles;
 - Price clubs: membership fee + per unit price
- **Price discrimination:** Different consumers pay different amounts depending on **observable** (or even **unobservable**) characteristics:
 - Senior citizen discounts;
 - Student discounts;
 - Saturday night stay on flights;
 - Quantity discounts
 - Personalized Prices

} We saw this
(group or 3rd degree)

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Sophisticated Pricing: Two Part Tariffs (2PTs)

- It would be great if we could do two things:
 1. Produce at q^* where $MC(q^*) = p(q^*)$ to eliminate dead-weight loss (remember, the demand function $p(q)$ is the marginal WTP!)
 2. Capture all the consumer surplus (total WTP!)
- It is “easy” to do the first thing:
 - Find q^* that solves $MC(q^*) = p(q^*)$ and charge $p^* = p(q^*)$
- It is “easy” to do the second thing **if consumers are homogeneous:**
 - Charge the whole consumer surplus of each consumer at q^* as a fixed fee, F^*
- Hence the two part tariff: (F^*, p^*) where p^* maximizes total surplus and F^* transfers it from the consumers to the firm!

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Example: People who love Chocolate

- Imagine the willingness to pay data was not 19 individual students, **but one person who really loves chocolate** (me!)
- Diminishing WTP (not me...): WTP for first piece is \$5, for 2nd piece is \$4.75, 3rd is \$4.5...
- Imagine there are N such consumers who are all the same.
- How can we create the highest surplus and extract it as profits?
- If we charge $p^* = MC = 1$ then each consumer will demand 14 (or 15) pieces.
- Then set F^* to **capture all the surplus** from each consumer:

$$F^* = (5 - 1) + (4.75 - 1) + (4.5 - 1) + \dots + (1.25 - 1) = \$30.75 !$$

5
4.75
4.5
4.25
4
3.75
3.25
3
2.75
2.5
2.25
2
1.5
1.25
1
0.75
0.5
0.25
0.1
0

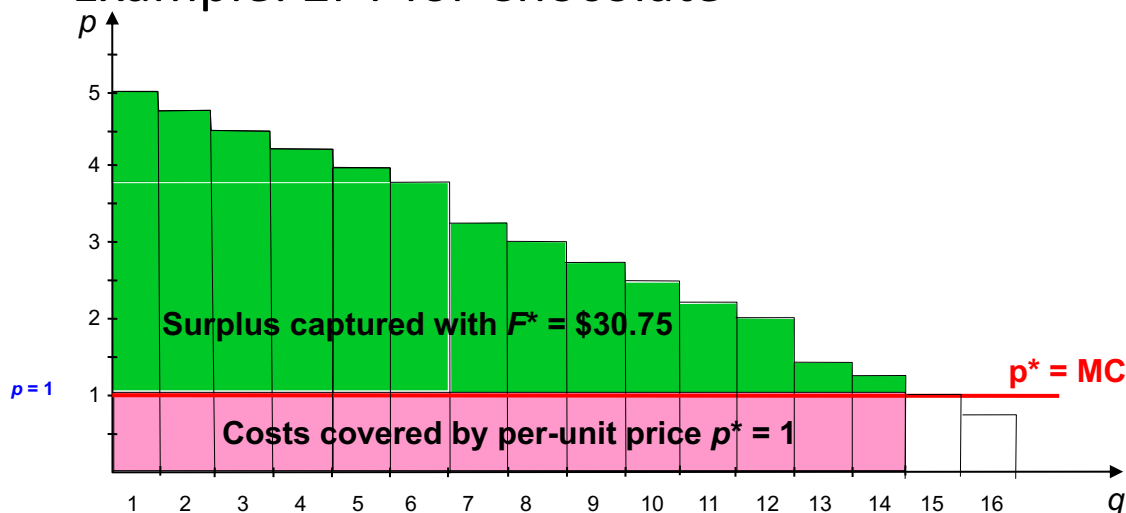
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Example: 2PT for Chocolate



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Two Part Tariffs with Heterogeneity

- Let's depart from our trivial world and move into a simple (yet far from trivial!) world:
 - Alice's demand for chocolate is $q_A = 5 - p$ ($p_A = 5 - q_A$)
 - Bob's demand for chocolate is $q_B = 4 - p$ ($p_B = 4 - q_B$)
 - Imagine $MC = 1$
- We'll look at three alternative:
 - Use Bob's preferences to determine (F^*, p^*) so Alice "smiles" (Bob's indifferent)
 - Use Alice's preferences to determine (F^*, p^*) so Bob is "priced out"
 - See if there is something better...

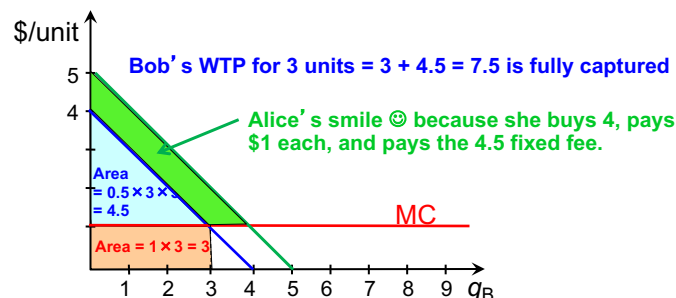
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Optimal 2PT for Bob's preferences



- Using Bob's demand:
 - Set $p^* = MC = 1$ to get $q_B = 3$
 - Charge Bob his remaining consumer surplus (Triangle): $F^* = 4.5$

$$\text{Total profits: } (4 + 3) \times (1 - 1) + (1 + 1) \times 4.5 = \$9$$

Alice

Bob

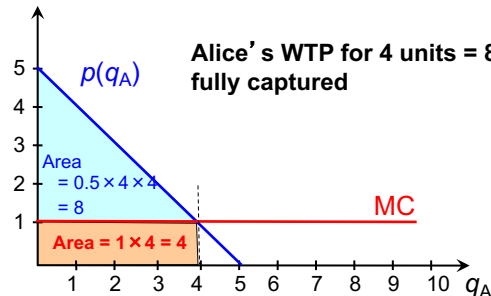
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Optimal 2PT for Alice's preferences



For any linear demand $p = a - bq$:
 Set price per-unit $p = MC$ to get optimal q^*
 Set $F^* = \frac{1}{2}(a - p)q^*$ (the surplus triangle)

- Using Alice's demand:
 - Set $p^* = MC = 1$ to get $q_A = 4$
 - Charge Alice remaining consumer surplus: $F^* = 8$
- Total profits are \$8 (Bob doesn't buy)
- It's better to charge Bob's 2PT!! (profits were \$9)

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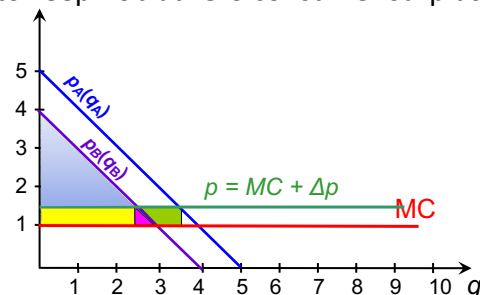
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Can we do better?

- Start with previous optimal solution: $p = \$1$ ($= MC$), $F = \$4.5$, $q_A = 4$, $q_B = 3$, profits = \$9
- Now raise p a bit (Δp) and reduce the fixed fee to keep Bob at zero consumer surplus
- Example: Set $p = 1.2 > MC$, $q_A = 3.8$, $q_B = 2.8$
 - Set F equal to Bob's remaining surplus:
 $F = 0.5 \times 2.8 \times 2.8 = 3.92 (< 4.5)$
 - Profits:
 $(3.8 + 2.8) \times 0.2 + 2 \times 3.92 = \$9.16 > \$9$
- General idea:
 - Reduce both types' gross CS (WTP) by a wee bit
 - Lose only a wee-wee bit on each Bob
 - Gain roughly $\Delta p \times (q_A - q_B)$ less a wee-wee bit on Alice!
- How big should Δp be? **Marginal Benefit = Marginal Loss!**



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Costco Is Cracking Down on Membership Sharing

1. What problem did self-checkout lanes introduce that interfered with Costco's pricing strategy?
2. How would you evaluate the costs and benefits of checking IDs at self-checkout versus eliminating self-checkout altogether?

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When 2-Part Tariffs Work: No Arbitrage

- Arbitrage is not a problem in many cases:
 - Amusement parks – you can't ride the rides for others ($F > 0, p = 0$)
 - Price Clubs – hard to buy for everyone
 - Ski resorts (same)
 - Cell phones – hard for you to supply everyone's phone service
 - Bars – you can't drink for others
 - Golf courses – you can't play for others
 - Costco – only undergrads do "Costco runs"
- These are situations where we tend to see two- (or multi-) part tariffs.

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Market Segmentation (2nd-Degree PD)

- There are several types of consumers, but you can't tell them apart
 - e.g., Student ID does not indicate WTP
 - Some people have a higher WTP than others – can't tell!
- Can't use several prices...
 - Everyone will buy at the cheapest price!
- Simple pricing won't do more than get at the aggregate demand
- What can we do to squeeze a bit more profits? (from CS and DWL)

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A Caricature: Start with Simple Pricing

- Consider the following example:
 - Alice's willingness to pay for one chocolate bar is \$5 and for 2 is \$8 (the 2nd is valued at 3, and she doesn't want more than 2).
 - Bob's willingness to pay for one chocolate bar is \$4.5 and for 2 is \$6 (the 2nd is valued at 1.5, and he doesn't want more than 2).
 - You're the seller, you have only variable costs $C(q) = q$ so $MC = 1$
- What can you do with simple pricing?
- You would only consider 4 prices:
 - $p = 5$: Alice buys 1, profits = $5 - 1 = 4$
 - $p = 4.5$: Alice & Bob both buy 1, profits = $9 - 2 = 7$ ← **Optimal!**
 - $p = 3$: Alice buys 2 & Bob buys 1, profits = $9 - 3 = 6$
 - $p = 1.5$: Alice & Bob both buy 2: profits = $6 - 4 = 2$

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A Caricature: Look at optimal 2-part-tariff

- What can you do with two-part tariffs?
 - Now here are two “types” of consumers... not that easy
 - Depending on who you want to sell to, you will need to set the fixed fee so that the person with the lowest surplus will be willing to pay it!
- You would only consider 2 two-part tariffs:
 - $(F = 1.5, p = 3)$: A buys 2 & B buys 1, profits = $2 \times 1.5 + 9 - 3 = 9$
 - $(F = 3, p = 1.5)$: A & B buy 2: profits = $2 \times 3 + 6 - 4 = 8$
- By selling 3 instead of 2 (simple price) we create more surplus, and reaped some of it (participation fee F) BUT chose not to create maximal surplus selling 4 (not homogeneous customers)
- What if Bob asks Alice to buy for him? (**Arbitrage!**)

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A Caricature: Quantity Discounts

- You can try to control arbitrage with **packaging** (quantity discounts):
 - You can charge a price for one bar, p_1
 - You can charge a price for **a pack** of two bars, p_2
 - **Assumption**: it's not easy to buy a two pack and split it. (And it may even not be a concern if one consumer wants more than his share of the split.)
- General idea:
 - Given p_1 and p_2 , one type of consumer may choose to buy only one bar, while another may choose to buy a 2-bar package
 - Need to find such prices that would maximize your profits.
 - (**NOTE**: simple pricing is a special case of quantity based pricing)
- What will determine which package each consumer buys?
- **Each will maximize net surplus!**

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A Caricature: Quantity Discounts

- How about selling in packages of 2 bars?
 - Case 1: Both buy 2 bar package (“new leaver”: play with unit size & price).
 - $p_2 = 6$ and don’t sell single units, make a profit of $2 \times 6 - 4 = \$8$
 - Case 2: Only Alice buys 2 bar package.
 - set $p_2 = 8$ and don’t sell single units, make a profit of $8 - 2 = \$6$
 - **Solution**: Case 1. (Note: Better than simple pricing (\$7) but not as good as the best two-part tariff (\$9).)
- Can we do better? Now things get tricky: You can have Alice buying 2 and Bob buying 1 (and vice versa). How do we price things?

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A Caricature: Quantity Discounts

- Case 3: Alice buys 2 and Bob buys 1.
 - Knee-jerk reaction: price each package to squeeze the juice out of the consumer buying it:
 - set price of 1 bar $p_1 = 4.5$ and set price of 2-bar package at $p_2 = 8$ (**quantity discount**), receive profits: $4.5 + 8 - 3 = 9.5$!
 - Are we cool?
 - NO! If Alice buys the 2-unit package, she gets **no consumer surplus**, but if she buys the single unit package, she gets 0.5 dollars worth of surplus
 - **Alice will buy the single bar at $p_1 = 4.5$ instead of the 2-bar-pack at $p_2 = 8$**
- The prices above violated **self selection** of consumers into the bundles we intended. If we can’t force it, the prices must make it happen!

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A Caricature: Quantity Discounts

- Case 4 done correctly: Alice buys 2 and Bob buys 1.
 - set price of 1 bar $p_1 = 4.5$ to squeeze all the juice from Bob (like 2PT)
 - set price of 2-bar package so that Alice (weakly) prefers to buy two units instead of 1. What's our constraint?
 - Alice can buy 1 unit and get a consumer surplus worth \$0.5.
 - We need to give Alice 0.5 worth of surplus on her 2-unit package.
 - Set at $p_2 = 7.5$, receive profits: $4.5 + 7.5 - 3 = \$9$!
- Just as good as two-part tariff! (not general...)
 - Because we get the same allocation: we extracted all of Bob's surplus from 1 unit, and again from Alice, together with her value of the second good.

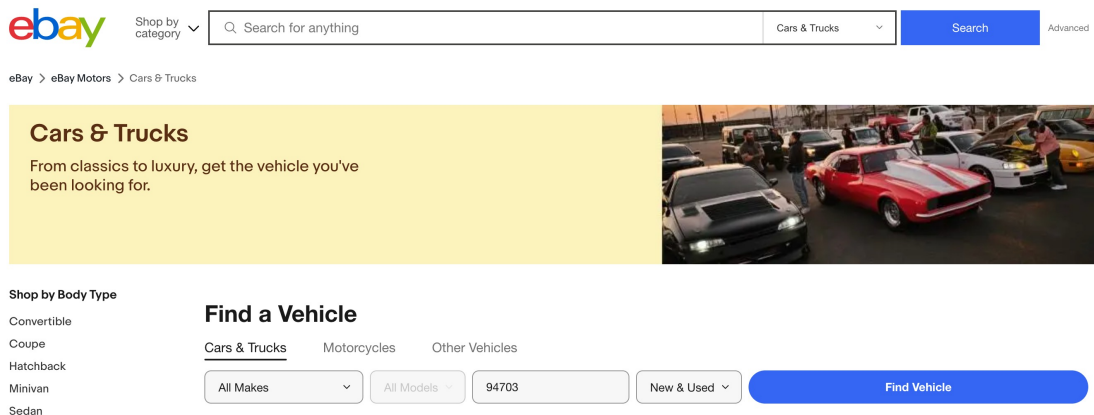
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Segmentation: Economic Intuition

- The optimal solution is interesting:
 - Alice gets 2 bars, and that is socially efficient! ($MC = 1 <$ than marginal $WTP = 3$)
 - Bob gets 1 bar, and that creates DWL! ($MC = 1 <$ than marginal $WTP = 1.5$)
- **Why?**
 - By making Bob's low quantity package smaller than what's efficient, it is not as good for him as the efficient size of 2 (he's 1.5 worse off gross)
 - But this makes Bob's low quantity package less desirable for Alice by 3 (gross)!
 - This in turn means that it's easier to squeeze more surplus out of Alice because her "cannibalizing" alternative (1-bar) is not very attractive for her.
- Cannibalization is a main concern in creating "packaging" (quantity discounts) because each consumer type has to have the incentive to buy what's tailored for them.

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Packaging (quantity discounts) in Reality



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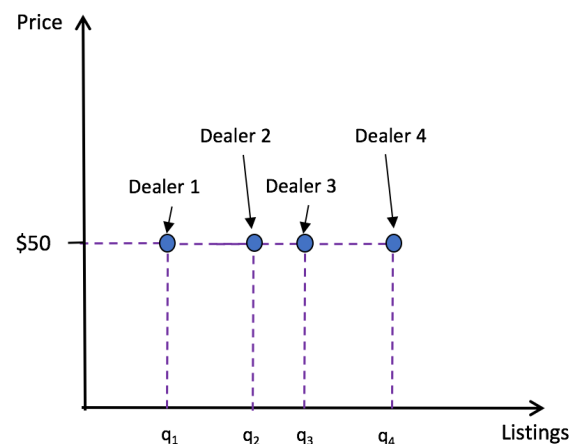
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eBay Motors – Real World Example

- eBay had a per-listing price for cars
- Idea: Subscription model with quantity discounts – great!
- Need WTP/demand data...
- ...but every dealer gave us one point on their demand curve for listings
- How do we get more data?
- Managerial incentives posed challenges...



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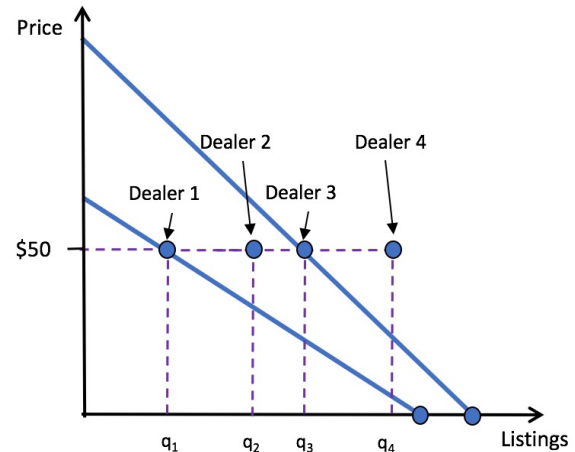
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eBay Motors: Eliciting Demand

- Offer flat fee, unlimited listings (2PT with $F > 0$ and $d p = 0$!)
- Calculate flat fee based on past 3 month listings
- Everyone is happy!
- Need to make assumptions on shape of demand...



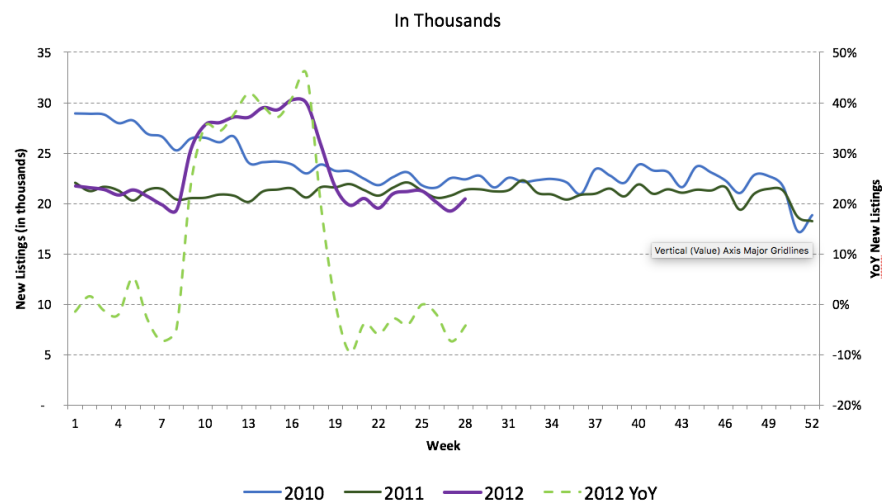
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Experiment 1: Listing Behavior



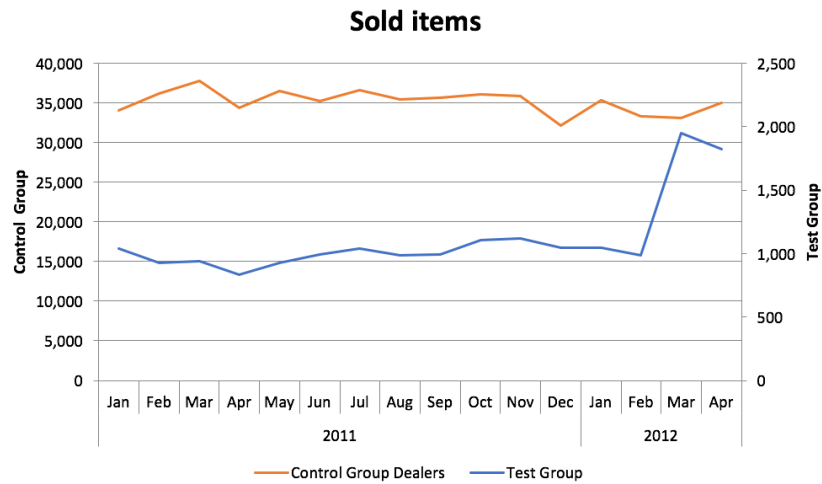
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Experiment 1: Sales



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Analysis (big version of Alices & Bobs)

"Nested" maximization program:

$$\max_{q_B, t_B, p_L} \sum_{i=1}^N (y_{iB} t_B + y_{iL} p_L q_{iL})$$

$$q_{iL} = \frac{p_L - a_i}{b_i}$$

$$s_{iL} = \frac{1}{2}(a_i - p_L)q_{iL}$$

$$s_{iB} = q_B p_{iB} + \frac{1}{2}(a_i - p_{iB})q_B - t_B$$

$$y_{iB} = \mathbf{1}(s_{iB} > s_{iL}, s_{iB} > 0)$$

$$y_{iL} = \mathbf{1}(s_{iL} > s_{iB})$$

$y_{iB} = 1$ if buyer i buys a package

$t_B =$ price of package

$y_{iL} = 1$ if buyer i buys single listings

$p_L =$ price per listing

$q_{iL} =$ number of listing of buyer i

$a_i - p_L =$ demand of buyer i for listings

$s_{iL} =$ buyer surplus if chooses listings

$s_{iB} =$ buyer surplus if chooses package

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Experiment 2: Sensitivity (demand shape...)

Version 1

- ▶ Tier 1: 10 monthly listings for \$450 per month
- ▶ Tier 2: 100 monthly listings for \$3,800 per month
- ▶ Tier 3: 200 monthly listings for \$6,600 per month

Version 2

- ▶ Tier 1: 10 monthly listings for \$450 per month
- ▶ Tier 2: 100 monthly listings for \$3,300 per month
- ▶ Tier 3: 200 monthly listings for \$5,000 per month

Version 3

- ▶ Tier 1: 10 monthly listings for \$430 per month
- ▶ Tier 2: 50 monthly listings for \$1,650 per month
- ▶ Tier 3: 100 monthly listings for \$2,800 per month

Version 4

- ▶ Tier 1: 10 monthly listings for \$430 per month
- ▶ Tier 2: 50 monthly listings for \$1,650 per month
- ▶ Tier 3: 100 monthly listings for \$2,800 per month
- ▶ Tier 4: 200 monthly listings for \$5,000 per month

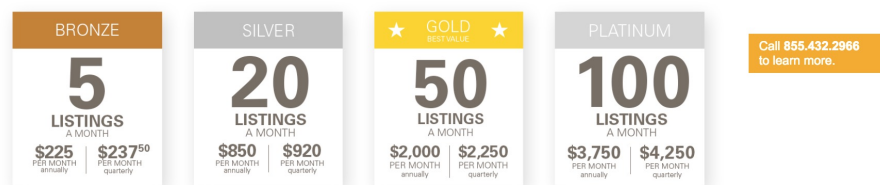
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Result: Launch New Program in 2014

Revenue increased by \$25MM in first year!

CHOOSE A PLAN THAT'S RIGHT FOR YOU.

Whether you're selling five or fifty vehicles, there's a right plan for your business, inventory, and budget. Simply choose from our four plan options, on an annual or quarterly cycle, and start streamlining your costs.[†]



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Summary

- Price discrimination (PD) helps reduce deadweight loss and increase profits
- In optimal Group-PD, MR's have to be equalized, and they will be equal to MC when there is no capacity constraint.
- Price discrimination transfer surplus to the monopolist; some consumers may be worse off while others might benefit.
- Two-part tariffs also reduce deadweight loss and increase profits
- Market Segmentation is a way to drive different types of consumers to different products that cause them to "self select" and reduces both consumer surplus and deadweight loss! (More on Tuesday)